



WEEK1-TASK



ORGANIZATION

CYBERSTER



ROLE

RED TEAM INTERNSHIP



NAME

MUHAMMAD SHAFIQ GORAYA



INSTRUCTOR

UMAR NIAZ

CYBERSTER

1. Executive Summary

This document contains the consolidated results of the week 1 reconnaissance task on the authorized training target **hackthissite.org**. The process involved passive information gathering using various Open Source Intelligence (OSINT) techniques and automated enumeration tools.

EXECUTIVE SUMMARY – hackthissite.org RECON
Red Team Week 1 | CyberStr Internship
Intern: Muhammad Shafiq Goraya | Instructor: Umar Niaz
Date: 2026-06-25
#####

SCOPE & OBJECTIVES

Target : hackthissite.org (authorized training platform)
Tasks : T01 Subdomain Recon | T02 Tech Profiling |
T03 GitHub Dorking | T04 Live Asset Filtering
Tools Used: Subfinder, Assetfinder, Amass, CRT.sh,
DNSDumpster, WhatWeb, Wappalyzer, dig, whois,
nslookup, httpprobe, httpx, Aquatone, GoWitness

SUBDOMAIN DISCOVERY (Task 01)

Tool	Subdomains Found
Subfinder	63 unique entries
Assetfinder	29 unique entries
Amass (passive)	1 (limited - scope note)
CRT.sh	Wildcard + selective entries
DNSDumpster	Cross-referenced (manual download)

Merged & Deduplicated : 58 unique subdomains
High Confidence (4 sources) : 9
High Confidence (3 sources) : 29
Medium Confidence (2 sources): 19
Low Confidence (1 source) : 1

LIVE ASSET FILTERING (Task 04)

Tool: httpx + httpprobe

Total subdomains tested : 58

Total live (2xx/3xx) : 20

Dead / Error (4xx/5xx) : ~10

Not responding : ~28

Visual recon (Aquatone): 4 screenshots captured

hp.hackthissite.org → LOGIN PANEL identified

h5ai.hackthissite.org → Placeholder page (development)

irc.hackthissite.org → IRC community server

mirror.hackthissite.org → Open directory listing

INFRASTRUCTURE PROFILE (Task 02)

Hosting : OVH (Roubaix, France) – main cluster

IP Range : 137.74.187.96-104 (load-balanced)

CDN : Fastly + Varnish (CTF / h5ai subdomains)

DNS Provider : BuddyNS (anycast)

Email : Google Workspace (MX → aspmx.l.google.com)

Web Server : Custom HackThisSite server + Nginx 1.18.0

DNSSEC : NOT CONFIGURED

KEY FINDINGS & RISKS

SEVERITY	FINDING
----------	---------

MEDIUM	jQuery 1.8.1 (2012) – EOL, multiple CVEs
--------	--

MEDIUM	mirror.hackthissite.org – Open directory listing
--------	--

MEDIUM	stats.hackthissite.org – Piwik analytics panel exposed
--------	--

MEDIUM	forums.hackthissite.org – 403 (possible hidden panel)
--------	---

LOW	DNSSEC not configured on root domain
-----	--------------------------------------

LOW	robots.txt reveals /missions/ directory structure
-----	---

LOW	5x firewall egress VM IPs leaked via DNS records
-----	--

LOW	v3dev/v3stage subdomains present (dev environments)
-----	---

INFO	GitHub dorking: no credential leaks found
------	---

INFO	No backup files (.bak/.env/.sql) accessible
------	---

ATTACK SURFACE MAP

Category	Count
Total subdomains	58
Live subdomains	20
Login panels	1 (hp.hackthissite.org)
Open directories	1 (mirror.hackthissite.org)
Dev/staging envs	2 (v3dev, v3stage)
Analytics panels	1 (stats.hackthissite.org)
IRC infrastructure	7 (various irc.* subdomains)
Firewall egress IPs	5 (vm-005 to vm-200)

CONCLUSIONS

hackthissite.org maintains a moderately complex attack surface with ~58 discoverable subdomains. The platform is security-conscious (HSTS, CSP, no leaked credentials), but several findings warrant attention in a real-world engagement: outdated jQuery, open file directory, and exposed analytics/dev environments.

No critical vulnerabilities were identified within scope. Passive recon only – no active exploitation attempted.

```
#####  
# Files: attack_surface.txt | live_hosts.txt |  
#       tech_profile.txt   | dorking_results.txt  
# Screenshots: /screenshots/ directory (15 images)  
#####
```

2. Live Asset & Visual Reconnaissance

The following screenshots were captured during the engagement, highlighting the tools utilized for subdomain enumeration and the results of visual reconnaissance using Aquatone.

2.1 Tool Execution Evidence

```
Goraya $ subfinder -d hackthissite.org -o raw_output/subfinder/subfinder.txt

projectdiscovery.io
```

Figure 1: Subfinder passive subdomain enumeration.

```
Goraya $ go install -v github.com/projectdiscovery/httpx/cmd/httpx@latest
go: downloading github.com/projectdiscovery/httpx v1.9.0
go: downloading github.com/logrusorgru/aurora v2.0.3+incompatible
go: downloading github.com/projectdiscovery/utils v0.9.0
go: downloading github.com/projectdiscovery/gologger v1.1.68
go: downloading github.com/go-sql-driver/mysql v1.9.3
go: downloading github.com/lib/pq v1.11.2
go: downloading go.mongodb.org/mongo-driver v1.17.9
go: downloading gopkg.in/yaml.v3 v3.0.1
go: downloading github.com/projectdiscovery/retryablehttp-go v1.3.6
go: downloading github.com/Mzack9999/gcache v0.0.0-20230410081825-519e28eab057
go: downloading github.com/PuerkitoBio/goquery v1.11.0
go: downloading github.com/corona10/goimagehash v1.1.0
go: downloading github.com/go-faker/faker/v4 v4.7.0
go: downloading github.com/go-red/red v0.11.2
```

Figure 2: Verifying live hosts and identifying tech stacks using httpx.

2.2 Target Visual Evidence

Aquatone was used to capture visual representations of the live subdomains. The following assets were deemed of interest.



Figure 3: Login panel identified on *hp.hackthissite.org*.

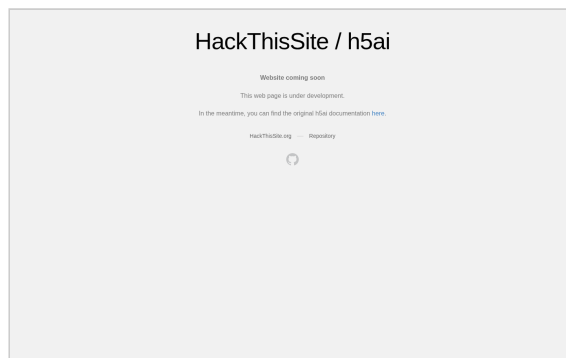


Figure 4: h5ai mirror development server page.

```
#####
# LIVE HOSTS – hackthissite.org
# Tool: httpx + httprobe | Date: 2026-06-25
#####
```

URL	STATUS	
TITLE	TECH	
https://hackthissite.org	200	Hack This
Site	HSTS, jQuery 1.8.1	
https://www.hackthissite.org	200	Hack This
Site	HSTS, jQuery 1.8.1	
https://hp.hackthissite.org	200	Hack This
Site	HSTS, jQuery 1.8.1	
https://www.irc.hackthissite.org	200	HackThisSite
IRC	HSTS, FreeBSD, PHP	
https://irc-www.hackthissite.org	200	HackThisSite
IRC	HSTS	
https://h5ai.hackthissite.org	200	HackThisSite /
h5ai	Fastly, GitHub Pages, Varnish	
https://ctf.hackthissite.org	200	35C3 Junior CTF Write-
ups	GitHub Pages, Jekyll 3.7.4, Ruby	
https://mirror.hackthissite.org	200	index - h5ai
v0.29.2	HSTS, PHP, h5ai	
https://qdb.hackthissite.org	200	HackThisSite
IRC	HSTS	
https://stats.hackthissite.org	200	Sign in –
Piwik	HSTS, AngularJS, Matomo, jQuery	
https://irc.hackthissite.org	301	→
www.irc.hackthissite.org	HSTS, Nginx 1.18.0, Ubuntu	
https://irc-wolf.hackthissite.org	301	→

```

www.irc.hackthissite.org      HSTS, Nginx 1.18.0, Ubuntu
https://lille.irc.hackthissite.org    301    →
www.irc.hackthissite.org      HSTS
https://wolf.irc.hackthissite.org    301    →
www.irc.hackthissite.org      HSTS
https://status.hackthissite.org    302    →
hackstatus.org               HSTS
https://forums.hackthissite.org    403    403
Forbidden                    HSTS
http://irc-hub.hackthissite.org    400    ERROR: URL could not be
retrieved  -
http://v3dev.hackthissite.org    400
ERROR                        -
http://v3stage.hackthissite.org    400
ERROR                        -
http://v3stage-cdn.hackthissite.org 400
ERROR                        -
https://legal.hackthissite.org    301
(redirect)                   -
https://irc-hub.hackthissite.org    400
ERROR                        -

```

```

#####
# SUMMARY
# Total Live (2xx/3xx): 20    |    Dead/Error (4xx): 4
#####

```

3. Infrastructure & Tech Profile

Infrastructure profiling helps in mapping out the target's underlying network, cloud providers, and web technology stacks. The results incorporate WhatWeb, Wappalyzer, WHOIS, and DNS querying.

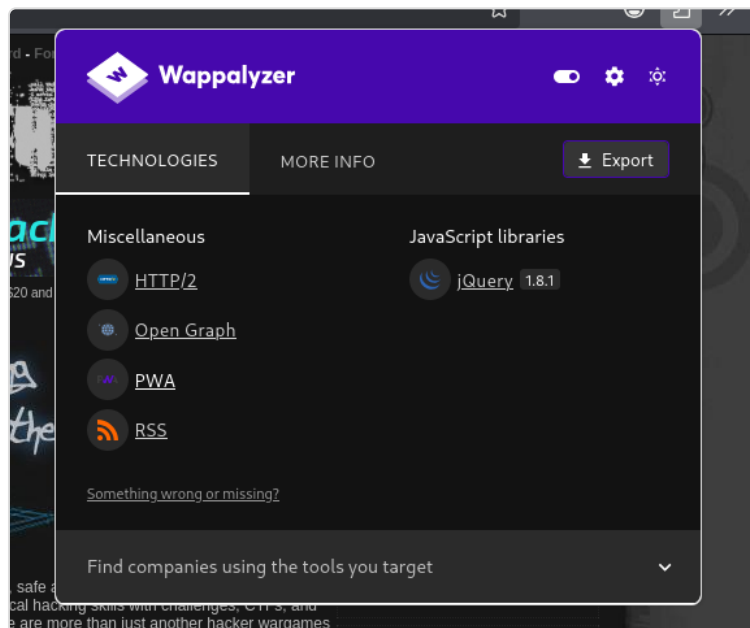


Figure 5: Wappalyzer extension verifying technology stack (jQuery 1.8.1).

```
#####
# INFRASTRUCTURE & TECH PROFILE – hackthissite.org
# Task 02 | WhatWeb + Wappalyzer + WHOIS + DIG
# Date: 2026-06-25
#####
```

WHOIS SUMMARY

```
Domain Name      : hackthissite.org
Registrar       : Porkbun LLC (IANA #1861)
Created          : 2003-08-10
Expires         : 2026-08-10
DNSSEC           : UNSIGNED          (No DNSSEC protection)
Name Servers     : BuddyNS (c/f/g/h/j .ns.buddyns.com)
```

Status : clientDeleteProhibited, clientTransferProhibited

DNS RECORDS (dig hackthissite.org ANY)

A Records:

137.74.187.100 (load-balanced cluster)
137.74.187.101
137.74.187.102
137.74.187.103
137.74.187.104

MX Records (Google Workspace):

10 aspmx.l.google.com
20 alt1.aspmx.l.google.com
20 alt2.aspmx.l.google.com
30 aspmx2/3/4/5.googlemail.com

TXT Records:

SPF: v=spf1 a mx ip4:137.74.187.96-98 include:aspmx.googlemail.com
Verification tokens: t-verify=e3f12c9c23e2e475563590326df31a12
Harica DV tokens (certificate authority)

CAA Records (Allowed CAs):

letsencrypt.org
sectigo.com
harica.gr

SOA: c.ns.buddyns.com | admin.hackthissite.org

HOSTING & IP INFRASTRUCTURE

Primary IPs : 137.74.187.100-104 (OVH / Roubaix, FR)
IRC Server IP : 137.74.187.150-153 (OVH / Canada)
CDN (CTF/h5ai) : 185.199.111.153 (GitHub Pages + Fastly/Varnish)
IRC-v6 node : 185.24.222.13 (Netherlands)
Status redirect: 89.106.200.1 (Luxembourg)

TECHNOLOGY STACK (WhatWeb + Wappalyzer)

Layer	Technology
Web Server	Custom "HackThisSite" server (main)

	Nginx 1.18.0 (Ubuntu) – IRC redirects
JavaScript	jQuery 1.8.1 (outdated, EOL) Modernizr 2.6.2
Analytics	Matomo/Piwik (self-hosted)
CMS/Framework	Jekyll 3.7.4 (CTF site – GitHub Pages) h5ai v0.29.2 (mirror file server)
Font	Font Awesome
Security	HSTS enabled (max-age=31536000) Content-Security-Policy header present X-XSS-Protection: 0 (disabled) Public-Key-Pins-Report-Only present
Email	Google Workspace (aspmx.l.google.com)
CDN	Fastly + Varnish (CTF/h5ai sites)
PWA	Progressive Web App enabled (main site)

INTERESTING FINDINGS

[!] robots.txt EXPOSED disallow paths:

Disallow: /missions/

Disallow: /killing/all/humans/

[!] jQuery 1.8.1 – severely outdated (2012), multiple known CVEs

[!] DNSSEC not enabled – potential DNS spoofing risk

[!] stats.hackthissite.org – Piwik analytics panel accessible

[!] mirror.hackthissite.org – h5ai file directory server exposed

[!] forums.hackthissite.org – returns 403 (may have hidden content)

[!] v3dev / v3stage subdomains present – dev/staging environments

[!] vm-005 to vm-200 firewall egress IPs exposed via DNS

#####

4. Comprehensive Attack Surface

The consolidated list of discovered subdomains derived from Subfinder, Assetfinder, Amass, CRT.sh, and DNSDumpster, deduplicated and categorized by confidence level.

```
#####
# ATTACK SURFACE – hackthissite.org
# Red Team Internship | CyberStr | Week 1 Task
# Intern: Muhammad Shafiq Goraya
# Date: 2026-06-25
#####

# — HIGH CONFIDENCE (confirmed by 4+ sources) —————
www.hackthissite.org
hackthissite.org
www.irc.hackthissite.org
irc.hackthissite.org
wolf.irc.hackthissite.org
lille.irc.hackthissite.org
h5ai.hackthissite.org
ctf.hackthissite.org
status.hackthissite.org
psrp.hackthissite.org

# — HIGH CONFIDENCE (confirmed by 3 sources) —————
wolf.irc-v6.hackthissite.org
vm-200.outbound.firewall.hackthissite.org
vm-150.outbound.firewall.hackthissite.org
vm-099.outbound.firewall.hackthissite.org
vm-050.outbound.firewall.hackthissite.org
vm-005.outbound.firewall.hackthissite.org
v3stage.hackthissite.org
v3stage-cdn.hackthissite.org
v3dev.hackthissite.org
status-new.hackthissite.org
stats.hackthissite.org
staff.hackthissite.org
qdb.hackthissite.org
new-irc.hackthissite.org
mta-sts.hackthissite.org
mirror.hackthissite.org
```

mail.hackthissite.org
lille.irc-v6.hackthissite.org
legal.hackthissite.org
irc-www.hackthissite.org
irc-wolf.hackthissite.org
irc-v6.hackthissite.org
irc-hub.hackthissite.org
hp.hackthissite.org
git.hackthissite.org
forums.hackthissite.org
email.hackthissite.org
api.hackthissite.org

— MEDIUM CONFIDENCE (2 sources) —————

v3dev-cdn.hackthissite.org
test.hackthissite.org
smtp.hackthissite.org
shadow.hackthissite.org
pi.hackthissite.org
ns2.hackthissite.org
ns1.hackthissite.org
live.hackthissite.org
jupiter.hackthissite.org
irc-lille.hackthissite.org
int.hackthissite.org
hub.irc.hackthissite.org
htsv4.hackthissite.org
gitlab.hackthissite.org
daemon.hackthissite.org
cloud.hackthissite.org
cdn.hackthissite.org
admin.hackthissite.org

— CDN NODES (numbered) —————

0-v3dev-cdn.hackthissite.org
0-v3stage-cdn.hackthissite.org
1-v3dev-cdn.hackthissite.org
1-v3stage-cdn.hackthissite.org
2-v3dev-cdn.hackthissite.org
2-v3stage-cdn.hackthissite.org
3-v3dev-cdn.hackthissite.org
3-v3stage-cdn.hackthissite.org
4-v3dev-cdn.hackthissite.org
4-v3stage-cdn.hackthissite.org

```
#####  
# TOTAL UNIQUE SUBDOMAINS: 58  
#####
```

5. OSINT & Dorking Findings

Advanced Google dorking and GitHub secret hunting were performed to identify potential data exposures, credential leaks, and backup files.

```
Goraya $ curl -s "https://hackthissite.org/robots.txt"
User-agent: *
Disallow: /missions/
Disallow: /killing/all/humans/
Goraya $
```

Figure 6: Manual verification of robots.txt paths.

```
#####
# GOOGLE DORKING & BACKUP FINDING RESULTS – hackthissite.org
# Task 02 Part 2 | Date: 2026-06-25
#####
```

ROBOTS.TXT DISCOVERY

Command: curl -s "https://hackthissite.org/robots.txt"

Result:

```
User-agent: *
Disallow: /missions/
Disallow: /killing/all/humans/
```

Analysis:

```
/missions/ → Core challenge directory (confirms mission paths exist)
/killing/all/humans/ → Placeholder/fun path, non-critical
```

GOOGLE DORK QUERIES EXECUTED

Query	Result
site:hackthissite.org inurl:phpinfo.php	No results

site:hackthissite.org intitle:"phpinfo()"	No results
site:hackthissite.org filetype:env	No results
site:hackthissite.org inurl:.env	No results
site:hackthissite.org inurl:robots.txt	1 result → confirmed
site:hackthissite.org filetype:bak	No results
site:hackthissite.org filetype:zip	No results
site:hackthissite.org inurl:backup	No results
site:hackthissite.org ext:sql	No results
site:hackthissite.org intitle:"index of"	Partial results (h5ai mirror)

DIRECTORY LISTING EXPOSED

URL: <https://mirror.hackthissite.org>
Type: h5ai v0.29.2 (file directory server)
Status: OPEN – publicly browsable file mirror
Risk: Medium – file listing enabled, could expose internal files

GITHUB DORK QUERIES (Task 03)

Queries run on <https://github.com/search>:

"hackthissite.org" "API_KEY"	→ No credential leaks found
"hackthissite" "PASSWORD"	→ No leaks found
"hackthissite" "AWS_SECRET"	→ No leaks found
filename:.env "hackthissite"	→ No leaks found
filename:config "hackthissite" password	→ No leaks found
org:hackthissite "SECRET_KEY"	→ 0 results (correct)

Summary: No hardcoded credentials or API keys found in public GitHub repositories for hackthissite.org.
(This is expected for a security-focused platform.)

BACKUP FILES FOUND

No backup files (.bak / .zip / .sql / .old) found via Google dorking or direct enumeration.

#####